

Multiple Timescales in Microbial Interactions

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 - Application: Antibiotic treatment

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“All models are wrong. . .”

Black-Scholes & the LTCM collapse (1998)

The model: Black-Scholes (1973) — prices options assuming log-normal returns, constant volatility, continuous markets.

The fund: Long-Term Capital Management, founded 1994.

- Board included Scholes & Merton (Nobel Prize 1997).
- Leverage $\sim 25:1$. Returns $> 40\%/yr$.

The failure: 1998 Russian debt crisis.

- Historical correlations broke simultaneously.
- Lost \$4.6B in 4 months $\rightarrow$ \$3.6B Fed-brokered bailout.

Lesson: A model's validity has a domain.
Outside it, leverage amplifies the error.

The point isn't "finance is risky" — it's that LTCM's models assumed stable historical correlations (and stable market conditions) and failed precisely when those correlations broke down (a lesson that applies equally to epidemiological models during novel outbreaks).

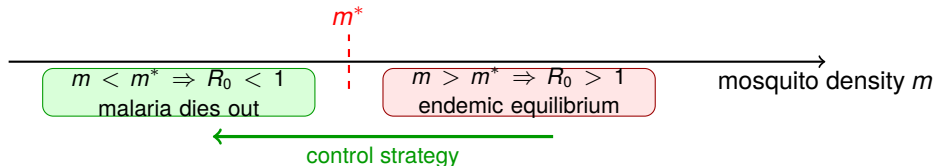
“...but some are useful.”

Ross's malaria model & the mosquito threshold (1908–1911)

The problem: Ross proved mosquitoes transmit malaria (Nobel 1902). But authorities doubted control was feasible: “*You can't kill every mosquito.*”

The model: Differential equations for human–mosquito transmission (“Theory of Happenings”, 1908).

The insight: A critical mosquito density m^* exists.



→ You don't need to kill *all* mosquitoes — just enough to cross the threshold.

This shaped malaria control campaigns worldwide and gave us R_0 .

Ross's model was enormously simplified — it ignored spatial structure, immunity, strain variation — but its core prediction, the threshold, was robust.

Why Study Mathematical & Epidemiological Modeling?

"All models are wrong, but some are useful."-George E. P. Box

"As a matter of fact all epidemiology, concerned as it is with the variation of disease from time to time or from place to place, must be considered mathematically, however many variables are implicated, if it is to be considered scientifically at all." - Sir Ronald Ross.

Beyond Intuition: Non-Linear Dynamics Why intuition fails in non-linear disease systems and how mathematical structure provides rigor.

- **The Limits of Verbal Reasoning** Biological systems involve non-linear feedback loops, delay mechanisms, and non-intuitive thresholds. Qualitative reasoning cannot reliably predict whether an outbreak will ignite, saturate, or oscillate over time.
- **The Power of Mathematical Rigor** Models convert qualitative hypotheses into precise equations. They force explicit definitions of assumptions, enable analytical stability analysis, and expose hidden systemic vulnerabilities.

Mechanistic Insight via Compartments

Compartmental frameworks (e.g., SIR, SEIR, SIS) disaggregate complex populations into state variables, tracking population flux using ODEs, for e.g., in the SIR model

$$\begin{cases} \frac{dS}{dt} = -\frac{\beta}{N}SI + \alpha R \\ \frac{dI}{dt} = \frac{\beta}{N}SI - \gamma I \\ \frac{dR}{dt} = \gamma I - \alpha R \end{cases}$$

with the total population N .

This allows us to quantify transmission rates β and recovery periods γ directly from biological mechanisms.

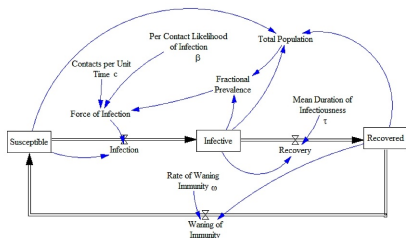


Figure 1: An example stock-flow diagram. This stock-flow diagram has three stocks labeled “Susceptible”, “Infective”, and “Recovered”; three flows labeled “Infection”, “Recovery”, and “Waning of Immunity”; many auxiliary variables including “Force of Infection” and “Total Population”; and links depicted by blue arrows

In-Silico Policy Experimentation & Uncovering Key Thresholds

- **Ethical Safety** We cannot perform trial-and-error experiments on human populations during lethal epidemics. Models allow virtual testing without biological harm.
- **Scenario Analysis** Compare non-pharmaceutical interventions (lockdowns, masking) against pharmaceutical strategies (vaccines, antivirals) under varying compliance rates.
- **Resource Allocation** Optimize limited health budgets by evaluating cost-effectiveness and distribution logistics prior to physical deployment.

Mathematical modeling introduced the **basic reproduction number** R_0 . If $R_0 > 1$, an infection invades; if $R_0 < 1$, it dies out.

From this single mathematical relation, we derive the critical herd immunity threshold required for global eradication $P_c = 1 - 1/R_0$.

*Highly recommended textbook: Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering - **Steven Strogatz**

From SIR to multi-strain co-infection

Level 1 (Classic SIR) 1 pathogen & 3 compartments.

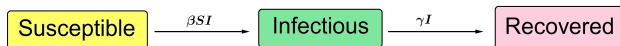


Figure 2: Standard SIR model. Dynamics governed by a single R_0 .

Level 2 (textbook-standard multi-strain): N competing strains, each with its own I_i , but a host carries at most one strain at a time.

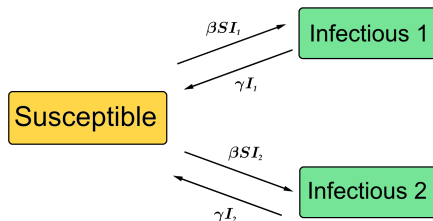


Figure 3: Standard multi-strain model. generically, the strain with the highest R_0 wins (competitive exclusion).

Multi-strain coinfection: a scheme for 2-strain system:

Level 3 (this talk): Allow a host already colonized by strain i to acquire strain j , creating co-colonized compartments I_{ij} .

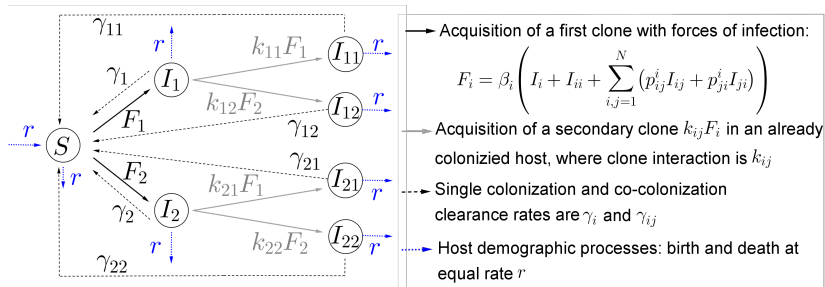


Figure 4: Model diagram when $N = 2$. This introduces new families of parameters: co-colonization susceptibilities k_{ij} , transmission probabilities from mixed hosts p_{ij}^s , and co-infection clearance rates γ_{ij} , and the system jumps from $N + 1$ to $N + N^2$ compartments. Competitive exclusion can break.

The challenge: With N strains, the full system has $O(N^2)$ equations. Direct analysis is intractable.

Motivation

Why co-infection-system? [van Baalen and Sabelis, 1995, Alizon, 2013]

- ① Multiple infections by parasite strain/ species are found everywhere in the wild.
- ② Multiple infections tend to worsen human health compared with single infections.
- ③ To reduce the complexity but allow us to analyze multiple-infections → co-infection

Note: to avoid giving a frequency-dependent advantage to the mutant → allow the coinfection by the same strain [van Baalen and Sabelis, 1995, Alizon, 2013].

SIS system with co-infection are complex to study

→ **Key assumption:** Strains are similar

In [Gjini and Madec, 2016, Madec and Gjini, 2020], authors study the case variation in co-colonization interaction coefficients (k_{ij})

We follow this work but complete for general case: variations in five traits

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The general model

- susceptible: S
- hosts colonized by strain i : I_i
- co-colonized hosts: strain i first and then strain 2: I_{ij} .

$$\begin{cases} \frac{dS}{dt} = r(1 - S) + \sum_{i=1}^N \gamma_i I_i + \sum_{i,j=1}^N \gamma_{ij} I_{ij} - S \sum_{i=1}^N F_i, \\ \frac{dI_i}{dt} = F_i S - (r + \gamma_i) I_i - I_i \sum_{j=1}^N k_{ij} F_j, & 1 \leq i \leq N, \\ \frac{dI_{ij}}{dt} = k_{ij} I_i F_j - (r + \gamma_{ij}) I_{ij}, & 1 \leq i, j \leq N \end{cases} \quad (2.1)$$

where

$$F_i = \beta_i \left(I_i + \sum_{j=1}^N (p_{ij}^i I_{ij} + p_{ji}^i I_{ji}) \right). \quad (2.2)$$

→ Assume: $S + \sum_{i=1}^N I_i + \sum_{i,j=1}^N I_{ij} = 1$ at all time.

Parameters and interpretations

- β_i : strain-specific transmission rates
- γ_i, γ_{ij} : clearance rates of single colonization/ co-colonization,
- p_{ij}^s : Transmission probability of $s \in \{i, j\}$; from a host co-colonized by strain- i then $-j$ (priority effects). $p_{ij}^i + p_{ij}^j = 1$
- k_{ij} : relative factor of altered susceptibility to co-colonization between colonizing strain i and co-colonizing strain j
- r : recruitment/ mortality rate, the same for all I_i and I_{ij} .

Setting $m = \gamma + r$.

Even in the similar strains, the parameter are not the same:

$$\begin{aligned} \beta_i &= \beta (1 + \epsilon b_i), & \gamma_i &= \gamma (1 + \epsilon \nu_i), & \gamma_{ij} &= \gamma (1 + \epsilon u_{ij}) \\ p_{ij}^s &= \frac{1}{2} + \epsilon \omega_{ij}^s, & k_{ij} &= k + \epsilon \alpha_{ij}. \end{aligned} \tag{2.3}$$

→ quasi-neutral assumption

New variables and Sketch of approach

Define new variables

- $I = \sum_{i=1}^N I_i$: total mass of single infected strains
- $D = \sum_{i,j=1}^N I_{ij}$: total mass of co-infected strains
- $J_i = I_i + \sum_{j=1}^N (p_{ij}^i I_{ij} + p_{ji}^i I_{ji})$. Then note that $F_i = \beta_i J_i$
- $T = I + D = \sum_{i=1}^N J_i$: total mass of infected and co-infected strains

Sketch of the approach

- 1 Consider the neutral system and approximate (S, T, I) by S^*, T^*, I^* .
- 2 Change the variables (I_i, J_i) to its weights (z_i, v_i) . Then rewrite the system using the ansatz and new variables.
- 3 Deduce the slow-fast model.
- 4 Using the Tikhonov's Theorem to deduce the replicator system for $(z_i)_{1 \leq i \leq N}$.

Neutral system

Taking $\epsilon = 0$ we have the *neutral system*

1

$$\begin{cases} \frac{dS}{dt} = r(1 - S) + \gamma T - S\beta T \\ \frac{dT}{dt} = \beta ST - mT \\ \frac{dI}{dt} = \beta TS - (m + k\beta T)I. \end{cases} \quad (2.4)$$

2

$$\begin{cases} \frac{dI_i}{dt} = \beta J_i S - mI_i - kI_i \beta T, \\ \frac{dJ_i}{dt} = (\beta S - m)J_i - \frac{1}{2}\beta kI_i T + \frac{1}{2}\beta kI_j J_i \end{cases} \quad (2.5)$$

3

$$\frac{dI_{ij}}{dt} = k\beta I_i J_j - mI_{ij}. \quad (2.6)$$

Analyzing on the neutral system

- ① We have $(S, T, I) \rightarrow (S^*, T^*, I^*)$ asymptotically as $t \rightarrow \infty$ with

$$R_0 = \frac{\beta}{m}, \quad S^* = \frac{1}{R_0}, \quad T^* = 1 - S^*, \quad I^* = \frac{mT^*}{m + \beta k T^*}, \quad D^* = T^* - I^* \quad (2.7)$$

- ② Pivot part $\rightarrow$ transferring to systems for (v_i, z_i)

- ③ Solved if (I_i, J_i) are known.

Note: R_0 is the basic reproductive number, i.e. average number of individuals infected from one other individual.

$R_0 < 1$: strain goes to extinction. $R_0 \geq 1$: strain survives.

Change the variables

- 1 Take $\epsilon = 0$ and substitute (S, T, I) by the limit (S^*, T^*, I^*) :

$$\frac{d}{dt} \begin{pmatrix} I_i \\ J_i \end{pmatrix} = \begin{pmatrix} -(m + \beta k T^*) & m \\ -\frac{\beta k T^*}{2} & \frac{\beta k I^*}{2} \end{pmatrix} \begin{pmatrix} I_i \\ J_i \end{pmatrix} = A \begin{pmatrix} I_i \\ J_i \end{pmatrix}. \quad (2.8)$$

- 2 Set $P = \begin{pmatrix} 2T^* & I^* \\ D^* & T^* \end{pmatrix}$ then $P^{-1} = \frac{1}{|P|} \begin{pmatrix} T^* & -I^* \\ -D^* & -2T^* \end{pmatrix}$ and

$$A = P \begin{pmatrix} -\xi & 0 \\ 0 & 0 \end{pmatrix} P^{-1}.$$

- 3 Set $\boxed{\begin{pmatrix} v_i \\ z_i \end{pmatrix} = P^{-1} \begin{pmatrix} I_i \\ J_i \end{pmatrix}}$ then $\begin{cases} \frac{dv_i}{dt} = -\xi v_i \\ \frac{dz_i}{dt} = 0 \end{cases}$

Then z_i is exactly the proportion prevalence of strain i in the total of infected mass.

Intuition: Why change variables?

The matrix A governing the neutral dynamics of (I_i, J_i) has eigenvalues $-\xi$ and 0. This tells us two things:

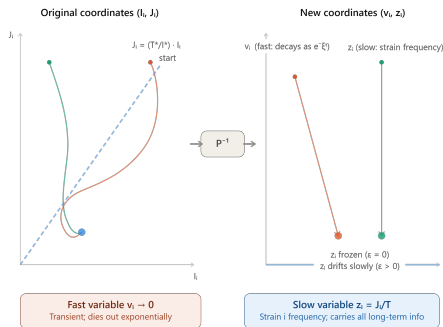


Figure 5: The change of variables via P^{-1} diagonalizes A and separates the dynamics into a transient part ($v_i \rightarrow 0$ fast) and the strain frequencies ($z_i = \text{const}$ when $\epsilon = 0$). When we turn on small trait differences ($\epsilon > 0$), only z_i acquires slow drift, giving us the replicator equation.

1. There's a fast direction

perturbations transverse to the slow manifold (the line $J_i = (T^*/I^*) \cdot I_i$) decay exponentially at rate ξ . This is the variable v_i .

2. There's a conserved direction

along the slow manifold, nothing changes in the neutral system. This is the variable $z_i = J_i/T^*$, which is exactly the frequency of strain i in the total infected population.

Theorem ([Tikhonov, 1952])

Consider the initial value problem

$$\begin{cases} \frac{dx}{d\tau} = f(x, y, \tau) + O(\epsilon), & x(0) = x_0, \quad x \in D \subset \mathbb{R}^n, \\ \epsilon \frac{dy}{d\tau} = g(x, y, \tau) + O(\epsilon), & y(0) = y_0, \quad y \in G \subset \mathbb{R}^m. \end{cases} \quad (2.9)$$

- a. We assume that a unique solution of the initial value problem exists and suppose this holds also for the reduced problem

$$\begin{cases} \frac{dx}{d\tau} = f(x, y, \tau), & x(0) = x_0, \\ 0 = g(x, y, \tau), \end{cases} \quad (2.10)$$

with solution $\bar{x}(\tau)$, $\bar{y}(\tau)$.

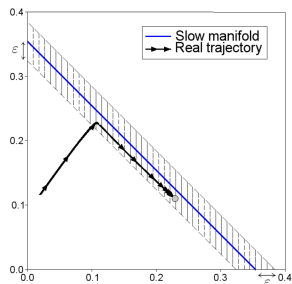


Figure 6: Slow manifold

- b. Suppose that $0 = g(x, y, \tau)$ is solved by $\bar{y} = \phi(x, \tau)$, where $\phi(x, \tau)$ is a continuous function and an isolated root. Also, *suppose that $\bar{y} = \phi(x, t)$ is an asymptotically stable solution of the equation $\frac{dy}{dt} = g(x, y, \tau)$, where $\tau = \epsilon t$, that is uniform in the parameters $x \in D$ and $t \in \mathbb{R}^+$.*
- c. $y(0)$ is contained in an interior subset of the domain of attraction of $\bar{y} = \phi(x, \tau)$ in the case of the parameter values $x = x(0)$, $\tau = 0$.

Then, we have

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} x_{\epsilon}(\tau) &= \bar{x}(\tau), & 0 \leq \tau \leq T, \\ \lim_{\epsilon \rightarrow 0} y_{\epsilon}(\tau) &= \bar{y}(\tau), & 0 < \tau_0 \leq \tau \leq T, \end{aligned} \tag{2.11}$$

with τ_0 and T are constants independent of ϵ .

An Approximation Theorem in Perturbation Theory

Theorem (Expansion Theorem, [Verhulst, 1989])

Consider the initial value problem

$$\frac{dx}{dt} = f_0(t, x) + \epsilon R(t, x, \epsilon), \quad x(t_0) = \eta, \quad (2.12)$$

and $|t - t_0| \leq h$, $x \in D \subset \mathbb{R}^n$, $0 \leq \epsilon \leq \epsilon_0$. Assume that:

- a. $f(t, x)$ continuous in t and x , 2 times continuously differentiable in x ;*
- b. $R(t, x, \epsilon)$ continuous in t, x and ϵ , Lipschitz-continuous in x .*

Let $x_0(t)$ be the solution of

$$\frac{dx}{dt} = f_0(t, x), \quad x_0(t_0) = \eta. \quad (2.13)$$

Let $T > 0$, there exist $C > 0$ (depending on T) such that $\forall \epsilon \in (0, \epsilon_0)$ and $\forall t \in (0, T)$, we have

$$\|x(t) - x_0(t)\| \leq C\epsilon \quad (2.14)$$

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Notations and ansatz

Notations: For $\chi_d \in \{0, 1\}$ for $d = 1, 2, 3, 4, 5$, denoting:

$$\begin{aligned} \beta_i &= \beta(1 + \chi_1 \epsilon b_i); & \gamma_i &= \gamma(1 + \chi_2 \epsilon \nu_i); & \gamma_{ij} &= \gamma(1 + \chi_3 \epsilon u_{ij}); \\ p_{ij}^s &= \frac{1}{2} + \chi_4 \epsilon \omega_{ij}^s \quad s \in \{i, j\} \quad (\omega_{ij}^i + \omega_{ij}^j = 0); & k_{ij} &= k + \chi_5 \epsilon \alpha_{ij}; \end{aligned} \quad (2.15)$$

Definition: To capture all the perturbations of order 1 in the equation of the $z_i \rightarrow$ ansatz:

$$\boxed{S = S^* + \epsilon X + O(\epsilon^2), \quad T = T^* - \epsilon X + O(\epsilon^2), \quad I = I^* + \epsilon Y + O(\epsilon^2).} \quad (2.16)$$

Setting: $L_i(t) = \frac{1}{2} \sum_{j=1}^N (u_{ij} l_{ij} + u_{ji} l_{ji}), \mathbf{z} = (z_1 \quad \dots \quad z_N), \mathbf{v} = (v_1 \quad \dots \quad v_N).$

Rewrite system using (v_i, z_i) and new time-scale

Change the **time scale** $\tau = \epsilon t$ and rewrite (2.1) in the slow motion τ :

$$\left\{ \begin{array}{l} \epsilon \frac{dX}{d\tau} = -\beta T^* X + \chi_1 F_X^1(\mathbf{v}, \mathbf{z}) + \chi_2 F_X^2(\mathbf{v}, \mathbf{z}) + \chi_3 F_X^3(\mathbf{L}) + O(\epsilon) \\ \epsilon \frac{dY}{d\tau} = \beta(S^* - T^* - kl^*)X - (m + \beta k T^*)Y + \chi_1 F_Y^1(\mathbf{v}, \mathbf{z}) + \chi_2 F_Y^2(\mathbf{v}, \mathbf{z}) \\ \quad + \chi_5 F_Y^5(\mathbf{v}, \mathbf{z}) + O(\epsilon) \\ \epsilon \frac{dL_i}{d\tau} = -mL_i + \chi_3 F_{L_i}(\mathbf{v}, \mathbf{z}) + O(\epsilon) \\ \epsilon \frac{dv_i}{d\tau} = -\xi v_i + O(\epsilon) \\ \frac{dz_i}{d\tau} = F_{z_i}(X, Y, \mathbf{L}, \mathbf{v}, \mathbf{z}) + O(\epsilon) \end{array} \right. \quad (2.17)$$

Fast variables: $y(\tau) = (X, Y, \mathbf{L}, \mathbf{v})$ and **Slow variable:** $x(\tau) = \mathbf{z}(\tau)$.

Apply the Tikhonov's Theorem

1st step: take $\epsilon = 0$ in (2.17) and show that the fast variable converge to an attractor $\phi(\mathbf{z})$ parametrized by the slow variable.

Lemma

Let $\epsilon = 0$ in (2.17). Then there exist a function $\Phi(\mathbf{z}) = (X^(\mathbf{z}), Y^*(\mathbf{z}), \chi_3 \mathbf{L}^*(\mathbf{z}), 0)$ such that the solution $(X, Y, \mathbf{L}, \mathbf{v}, \mathbf{z})$ of (2.17) in fast times scale t , with any initial condition $(X, Y, \mathbf{L}, \mathbf{v}, \mathbf{z})(0)$ verifies $\mathbf{z}(t) = \mathbf{z}_0$ for all $t \geq 0$ and*

$$\lim_{t \rightarrow +\infty} (X, Y, \mathbf{L}, \mathbf{v})(t) = \Phi(\mathbf{z}_0).$$

Moreover, X^ and Y^* are linear function of the χ_i .*

Proof.

Sketch: Using the triangular structure of (2.17) the idea is to compute the limits step by step of $\mathbf{v}$, $\mathbf{L}$, X and Y in this order. □

A lemma to linearly combine

2nd step: take $\epsilon = 0$ in (2.17) and we fix

$$(X, Y, \mathbf{L}, \mathbf{v})(\tau) = \Phi(\mathbf{z}(\tau)). \quad (2.18)$$

3rd step: N last equations give the *slow system*

$$\frac{dz_i}{d\tau} = F_{z_i}(X^*(\mathbf{z}), Y^*(\mathbf{z}), \mathbf{L}^*(\mathbf{z}), 0, \mathbf{z}). \quad (2.19)$$

Lemma

Let $\mathcal{A} \subset \{1, \dots, 5\}$. Recall that $\chi_d = 1$ if $d \in \mathcal{A}$ and $\chi_d = 0$ if $d \notin \mathcal{A}$. The functions F_{z_i} for $i = 1, \dots, N$ in (2.19) read

$$F_{z_i}(X^*(\mathbf{z}), Y^*(\mathbf{z}), \mathbf{L}^*(\mathbf{z}), 0, \mathbf{z}) = \sum_{d=1}^5 \chi_d z_i f_{z_i}^d(\mathbf{z}),$$

where the functions $f_{z_i}^d$ do not depend on χ_d . In particular, if $\mathcal{A} = \{d\}$ for some $d \in \{1, 2, 3, 4, 5\}$, then $F_{z_i}(X^*(\mathbf{z}), Y^*(\mathbf{z}), \mathbf{L}^*(\mathbf{z}), 0, \mathbf{z}) = z_i f_{z_i}^d(\mathbf{z})$.

Define $\Theta_1 = \chi_1 \frac{2\beta S^* T^{*2}}{|P|}$, $\Theta_2 = \chi_2 \frac{\gamma I^* (I^* + T^*)}{|P|}$, $\Theta_3 = \chi_3 \frac{\gamma T^* D^*}{|P|}$, $\Theta_4 = \chi_4 \frac{2m T^* D^*}{|P|}$ and $\Theta_5 = \chi_5 \frac{\beta T^* I^* D^*}{|P|}$ and

$$\Theta = \Theta_1 + \Theta_2 + \Theta_3 + \Theta_4 + \Theta_5 \quad \text{and} \quad \theta_i = \frac{\Theta_i}{\Theta}. \quad (2.20)$$

Theorem (Main Result, [Le et al., 2021a])

Consider the system of equations

$$\begin{cases} \dot{z}_i = \Theta z_i ((\Lambda \mathbf{z})_i - \mathbf{z}^T \Lambda \mathbf{z}), & i = 1, \dots, N, \\ z_1 + z_2 + \dots + z_N = 1. \end{cases} \quad (2.21)$$

where Λ is the square matrix of size $N \times N$ whose coefficients $(i; j)$ are the pairwise invasion fitnesses λ_{ij}^j which satisfy, with $\mu = \frac{I^*}{D^*}$,

$$\begin{aligned} \lambda_{ij}^j = & \theta_1 (b_i - b_j) + \theta_2 (-\nu_i + \nu_j) + \theta_3 (-u_{ij} - u_{ji} + 2u_{jj}) \\ & + \theta_4 (\omega_{ij}^i - \omega_{ji}^j) + \theta_5 (\mu (\alpha_{ji} - \alpha_{ij}) + \alpha_{ji} - \alpha_{jj}). \end{aligned} \quad (2.22)$$

Then, for any initial values of (2.1), for each $\tau_0 > 0$, $T > \tau_0$ arbitrarily and independent on ϵ , there is $\epsilon_0 > 0$, $C > 0$ and a vector of positive coefficients $\mathbf{z}_0 \in \mathbb{R}^N$ verifying $\sum_{i=1}^N \mathbf{z}_{0,i} = 1$, such that $\forall \epsilon < \epsilon_0$

$$\left| S\left(\frac{\tau}{\epsilon}\right) - S^* \right| + \sum_{i=1}^N \left| I_i\left(\frac{\tau}{\epsilon}\right) - I^* z_i(\tau) \right| + \sum_{i,j=1}^N \left| I_{ij}\left(\frac{\tau}{\epsilon}\right) - D^* z_i(\tau) z_j(\tau) \right| \leq \epsilon C \quad (2.23)$$

$\forall \tau \in (\tau_0, T)$, where (S, I_i, I_{ij}) is the solution of (2.1) and $(z_1, \dots, z_N)$ is the solution of reduced system (2.21) together with $\mathbf{z}(0) = \mathbf{z}_0$.

Sketch of proof:

- ① Use Tikhonov's Theorem: approximate solutions of (2.17) using (2.19).
- ② Explicit computations (each single case) & Lemma 4: (2.19) $\iff$ (2.21)
- ③ By Step 1 & Expansion Theorem 2:

$$\left| S\left(\frac{\tau}{\epsilon}\right) - S^* \right| + \sum_{i=1}^N \left| I^* z_i(\tau) - I_i\left(\frac{\tau}{\epsilon}\right) \right| + \sum_{i=1}^N \left| T^* z_i(\tau) - J_i\left(\frac{\tau}{\epsilon}\right) \right| \leq \epsilon C_T$$
- ④ By Step 3 & Expansion Theorem 2:
$$\sum_{i,j=1}^N \left| I_{ij}\left(\frac{\tau}{\epsilon}\right) - D^* z_i(\tau) z_j(\tau) \right| \leq \epsilon C_T$$



Table 1: Summary of multi-strain outcomes for each case of a single trait varying Note that only the ratio between the Θ_i impact the qualitative behavior.

Trait varying	Type of dynamics
Transmission rates β_i	Competitive exclusion
Single infection clearance rates γ_i	Competitive exclusion
Co-infection clearance rates γ_{ij}	Possibility of Coexistence
Transmission probability from mixed carriage p_{ij}^S	Family of cycles
Co-colonization interaction factor via altered susceptibilities, k_{ij} [Gjini and Madec, 2016]	Anything

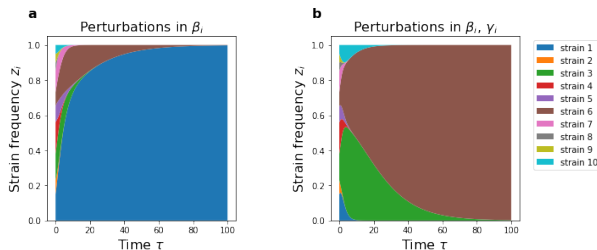


Figure 7: Illustration of competitive exclusion dynamics for $N = 10$, perturbations in transmission and clearance rates. (a) the strain with the highest reproduction number will persist while all other strains will go extinct. (b) strain 10 has highest basic reproduction number but strain 6 is the ultimate winner.

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Recalling that:

- ① λ_i^j : invasion fitness of strain i in an equilibrium set by strain j alone.
- ② $z^* = (z_1^*, z_2^*) = \frac{1}{\lambda_1^2 + \lambda_2^1} \begin{pmatrix} \lambda_1^2 \\ \lambda_2^1 \end{pmatrix}$ be the nonzero equilibrium state of (2.21) when $N = 2$.
- ③ Conditions on λ_1^2 and λ_2^1 for four scenarios

Table 2: System equilibria for 2-strain dynamics according to λ_1^2 and λ_2^1

	$\lambda_2^1 > 0$	$\lambda_2^1 < 0$
$\lambda_1^2 > 0$	Stable coexistence	Exclusion of strain 1
$\lambda_1^2 < 0$	Exclusion of strain 2	Bistable (unstable coexistence)

We have some remarks:

- If variations are in all traits, when $\mu = \frac{I^*}{D^*} \rightarrow \infty$, long time behavior is competitive exclusion. The case variations in only k_{ij} is proved in [Gjini and Madec, 2021]
- Variations in only β_i , γ_i and $p_{ij}^s \rightarrow$ exclusion
- Perturbations in γ_{ij} or k_{ij} can break the competitive exclusion

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Depending on how many traits and which traits vary in the system, we can have at most one, two, or more shifts with μ .

- Variation only in 2 strains only, can be in transmission rates β_i , clearance rates γ_i and transmission probability from co-colonized hosts $p_{ij}^s \rightarrow$ at most one shift in final outcome as a function of μ .
- Necessary condition for the presenting of 4 survival outcomes: variations are in coinfection clearance rates γ_{ij} and co-colonization interaction k_{ij}

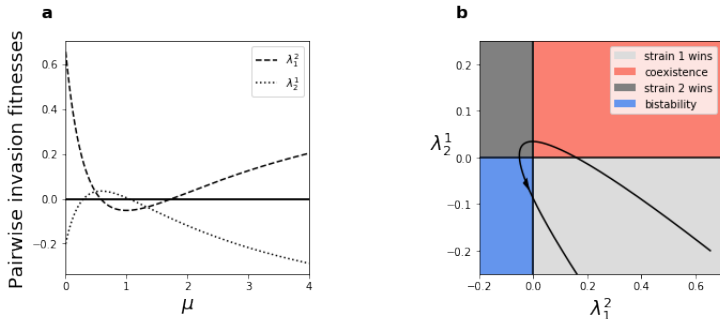


Figure 8: Four ecological scenarios may happen depending on μ (single-to-co-infection ratio) under fixed trait variation, [Le et al., 2021b]. Variations in transmission rates, infection and coinfection clearance rates, and co-colonization susceptibilities between strains.

With only values of basic reproductive number $R_{0,1}$ and $R_{0,2}$, it can not determine the long time behavior of the dynamics [Le et al., 2021b].

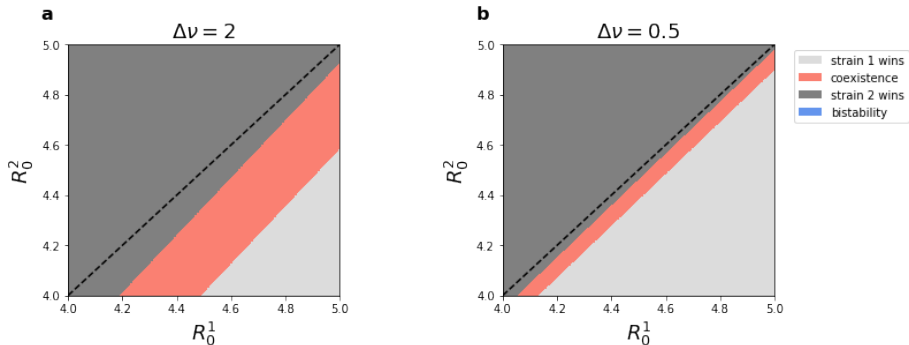


Figure 9: Scenarios according to basic reproductive numbers of strains $R_{0,1}$ and $R_{0,2}$ when $k = 3$, $m = 1.5$, $\beta = 6$, $\epsilon = 0.1$. We vary Δb in each case to obtain the relative comparison between $R_{0,1}$ and $R_{0,2}$ shown in the figures. The variations are in transmission rates β_i , single clearance rates γ_i , and for coinfection clearance rate we assume γ_{ij} with $u_{12} = u_{21} = \min \{\nu_1, \nu_2\}$. Figure (a) simulates when $\nu_1 = -1$, $\nu_2 = 1$. Figure (b) simulates when $\nu_1 = -0.25$, $\nu_2 = 0.25$.

Example

Illustrations of possible outcomes as a function of relative variation in transmission and clearance rate between two strains, for different values of k and R_0 , with $\gamma = 1$, $r = 0.2$.

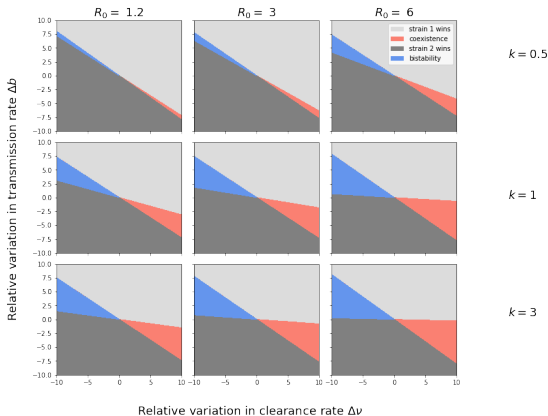


Figure 10: Perturbations only β_i , γ_i and γ_{ij} with $u_{11} = u_{12} = u_{21} = \nu_1$ and $u_{22} = \nu_2$

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- Suppose strain 1 is antibiotic-sensitive and strain 2 is antibiotic resistant.
- Antibiotic resistance usually entails a fitness cost $\rightarrow$ assume $\Delta b > 0$, $\Delta \nu > 0$.

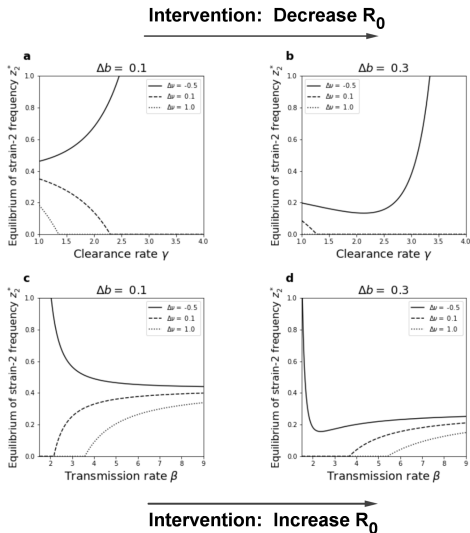


Figure 11: Increasing global clearance or global transmission can have opposite effects on single strain frequency at the endemic equilibrium, depending on underlying trait variation.

References



Alizon, S. (2013).

Co-infection and super-infection models in evolutionary epidemiology.
Interface Focus, 3.



Gjini, E. and Madec, S. (2016).

A slow-fast dynamic decomposition links neutral and non-neutral coexistence in interacting multi-strain pathogens.
Theoretical Ecology, 10:129–141.



Gjini, E. and Madec, S. (2021).

The ratio of single to co-colonization is key to complexity in interacting systems with multiple strains.
Ecology and Evolution, 11:8456 – 8474.



Le, T. M. T., Gjini, E., and Madec, S. (2021a).

Quasi-neutral dynamics in a coinfection system with n strains and asymmetries along multiple traits.
arXiv: Dynamical Systems.

 Le, T. M. T., Madec, S., and Gjini, E. (2021b).

Disentangling how multiple traits drive 2 strain frequencies in sis dynamics with coinfection.

bioRxiv.

 Madec, S. and Gjini, E. (2020).

Predicting n-strain coexistence from co-colonization interactions: Epidemiology meets ecology and the replicator equation.

Bulletin of Mathematical Biology, 82:1–26.

 Tikhonov, A. (1952).

Systems of differential equations containing a small parameter multiplying the derivative.

Mat. SB., 31:575–586.

 van Baalen, M. and Sabelis, M. W. (1995).

The dynamics of multiple infection and the evolution of virulence.

The American Naturalist, 146:881 – 910.

 Verhulst, F. (1989).

Nonlinear differential equations and dynamical systems.

Springer.